

Extended Summary — Matrices, Determinants, Linear Systems & Lines in the Plane

(Fusing "Linear Systems of Equations" with the Matrices/Determinants/Lines-in-the-plane parts of "Elements of Linear Algebra and Geometry")

1. Matrices

1.1 Concept of a matrix

A matrix is a rectangular table of $m \times n$ numbers arranged in m rows and n columns, written $A = (a_{ij})$, where a_{ij} is the element in row i , column j . Its **dimension** is $m \times n$. Matrices give a compact way to represent a system of linear equations (coefficients + constants) or any other tabular data (e.g., a set of vectors written as columns).

Example: $A = [[3, -1, 2], [4, 2, -5]]$ has dimension 2×3 ; $a_{23} = -5$, $a_{12} = -1$.

Two matrices are **equal** only if they have the same dimension and identical entries in every position.

Transposed matrix A^T : obtained by turning rows into columns. $(A^T)^T = A$.

1.2 Special matrices

- **Row/column matrix:** only 1 row / 1 column.
- **Null matrix:** all entries 0.
- **Square matrix of order n :** $m = n$; its **main diagonal** is the set $\{a_{11}, a_{22}, \dots, a_{nn}\}$.
- **Symmetric matrix:** $A = A^T$ ($a_{ij} = a_{ji}$).
- **Diagonal matrix:** every off-diagonal entry is 0.
- **Identity matrix I_n :** diagonal matrix with 1's on the diagonal.
- **Upper/lower triangular matrix:** all entries below/above the diagonal are 0.

1.3 Operations with matrices

- **Sum/subtraction:** same dimension required; add entry-by-entry. Associative, commutative, has an identity (the null matrix) and an inverse ($-A$).
- **Scalar multiplication:** $k \cdot A$ multiplies every entry by k .
- **Matrix product:** $A \cdot B$ is defined only if the number of columns of A equals the number of rows of B ; then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$. It is associative and distributive over addition, but **not commutative** ($A \cdot B \neq B \cdot A$ in general). If A is square of order n , $A \cdot I_n = I_n \cdot A = A$.
- **Inverse matrix A^{-1} :** if it exists, $A \cdot A^{-1} = A^{-1} \cdot A = I_n$. A matrix without an inverse is called **singular**.

Example: If $A = [[a, b, c], [d, e, f]]$ and the constant matrix is added: entries add position by position; product example: multiplying a 2×3 by a 3×3 gives a 2×3 result, only possible

because columns of A (3) match rows of B (3).

2. Determinants

2.1 Order 2 and 3

Every square matrix A has an associated number, $\det(A)$ or $|A|$.

- Order 1: $|a_{11}| = a_{11}$.
- Order 2: $|A| = a_{11}a_{22} - a_{12}a_{21}$.
- Order 3 (**Sarrus' rule**): $|A| = a_{11}a_{22}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{23}a_{12} - a_{13}a_{22}a_{31} - a_{23}a_{32}a_{11} - a_{33}a_{21}a_{12}$.

Example: $|[[1,-2],[0,3]]| = 1 \cdot 3 - 0 \cdot (-2) = 3$.

2.2 Order 4 or higher: adjugates (cofactors)

The **adjugate (cofactor)** A_{ij} of element a_{ij} is: (a) the determinant left after deleting row i and column j (the **minor**), with (b) sign $(-1)^{i+j}$ attached (a checkerboard pattern of $+/-$ starting with $+$ at position 1,1).

Cofactor expansion: the determinant can be computed along any row i or column j: $|A| = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$ (expanding by row i), or the analogous formula by column j. It is most efficient to expand along the row/column with the most zeros.

Example: expanding a 4×4 determinant along its second row (which has three zeros) reduces the work to computing a single 3×3 adjugate.

2.3 Properties of determinants

1. $\det(A) = \det(A^T)$.
2. Swapping two rows (or columns) flips the sign of the determinant.
3. Two equal or proportional rows/columns $\Rightarrow$ determinant is 0.
4. A null row or column $\Rightarrow$ determinant is 0.
5. Multiplying one row/column by k multiplies the whole determinant by k.
6. If a row/column is written as a sum, the determinant splits into the sum of two determinants (one per term).
7. If a row/column is a linear combination of others, the determinant is 0.
8. Adding a linear combination of other rows/columns to one row/column doesn't change the determinant (this is the basis of Gaussian-elimination-style simplification of determinants).
9. $\det(A \cdot B) = \det(A) \cdot \det(B)$.

2.4 Computing the inverse via the adjugate

Proposition: A square matrix A has an inverse if and only if $\det(A) \neq 0$ (such a matrix is called **regular** or **non-singular**). In that case:

$$A^{-1} = \text{adj}(A)^T / \det(A)$$

where $\text{adj}(A)$ is the matrix obtained by replacing every entry of A with its adjugate (cofactor).

Example: for $A = \begin{bmatrix} 2 & 0 & 1 \\ -2 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $|A| = -4 \neq 0$, so A is invertible; computing all nine adjugates, assembling $\text{adj}(A)$, transposing, and dividing by -4 gives A^{-1} , and one can check $A \cdot A^{-1} = A^{-1} \cdot A = I_3$.

2.5 Rank of a matrix (via determinants)

The **rank** of a matrix, $\text{rk}(A)$, is the number of linearly independent rows (equivalently, columns).

- A **minor of order h** is the determinant obtained by selecting h rows and h columns from A .
- A **bordered minor** is obtained by adding one more row and one more column (taken from the original matrix, at positions aligned with the existing minor) to a given minor.

Rank theorem: the rank of a (non-null) matrix equals the order of its largest non-zero minor.

Practical method: find a non-zero minor of order h ; then check its bordered minors of order $h+1$. If all of them are 0, the rank is h ; if one is non-zero, repeat the process with that larger minor.

Example: for a 3×4 matrix, if a 2×2 minor is non-zero but every bordered 3×3 minor built from it is 0, the rank is 2.

This same idea gives a determinant test for **linear independence**: k vectors in $\mathbb{R}^n$, arranged as rows/columns of a matrix A , are linearly independent iff $\text{rk}(A) = k$; in particular, n vectors in $\mathbb{R}^n$ are independent iff $\det(A) \neq 0$.

3. Systems of Linear Equations (SLE)

3.1 Definition and matrix notation

A system of m linear equations in n unknowns: $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1, \dots, a_{m1}x_1 + \dots + a_{mn}x_n = b_m$

can be written as $A \cdot X = B$, where A ($m \times n$) is the **coefficient matrix**, X ($n \times 1$) the vector of unknowns, and B ($m \times 1$) the **constant vector**. Concatenating A and B gives the **augmented matrix** $M = (A|B)$.

Two systems are **equivalent** if they share the same solution set (even with a different number of equations).

3.2 Classification of systems

- **Consistent:** has at least one solution.
 - **Independent:** exactly one solution.

- **Dependent:** infinitely many solutions.
- **Inconsistent:** no solution.

3.3 Rouché–Capelli theorem

Given m equations, n unknowns:

- $\text{rk}(A) = \text{rk}(M) = n \Rightarrow$ **independent** (unique solution).
- $\text{rk}(A) = \text{rk}(M) = r < n \Rightarrow$ **dependent**, with $(n - r)$ **degrees of freedom**.
- $\text{rk}(A) < \text{rk}(M) \Rightarrow$ **inconsistent**.

Example: for a 3×3 system, if $|A| \neq 0$ the system is automatically independent; if $|A| = 0$ you must compare $\text{rk}(A)$ with $\text{rk}(M)$ using minors to decide between dependent and inconsistent.

3.4 Homogeneous systems

If every constant term is 0 ($A \cdot X = 0$), the augmented matrix trivially has the same rank as A , so the system is **always consistent**: it has at least the **trivial solution** $x_1 = x_2 = \dots = x_n = 0$. It has *other* (non-trivial) solutions exactly when $\text{rk}(A) < n$.

Example: a 3×3 homogeneous system with $|A| \neq 0$ has only the trivial solution; forcing $|A| = 0$ (by choosing parameters) is what opens the door to non-trivial solutions.

4. Solving Methods

4.1 Gaussian elimination

Perform row operations (swap rows, scale a row, add a multiple of one row to another, remove an all-zero row) on the augmented matrix M until it reaches **row-echelon form** (a "staircase" of leading entries). Then solve by **back-substitution**, starting from the last non-trivial equation and working upward, expressing variables in terms of any remaining free parameters. The number of non-zero rows in the echelon form is precisely $\text{rk}(M)$.

4.2 Cramer's rule

Applies when A is **square** with $\det(A) \neq 0$ (a **Cramer's system**): the system is then guaranteed consistent and independent, and

$$x_i = \det(A \text{ with column } i \text{ replaced by } B) / \det(A), \text{ for } i = 1, \dots, n.$$

If the system is consistent-dependent ($\text{rk } r < n$), a Cramer's system can still be recovered: keep only r independent equations and r variables tied to a non-zero $r \times r$ minor, treating the remaining $(n - r)$ variables as free parameters; then apply Cramer's rule for each fixed value of those parameters.

5. Equations of a Line in the Plane

Given two points $P(p_1, p_2)$ and $Q(q_1, q_2)$ on a line r , the **director vector** is $v = PQ = (v_1, v_2) = (q_1 - p_1, q_2 - p_2)$. Any point $X(x, y)$ on the line satisfies:

1. **Vectorial equation:** $X = P + k \cdot v$, $k \in \mathbb{R}$.
2. **Parametric equations:** $x = p_1 + k \cdot v_1$, $y = p_2 + k \cdot v_2$ ($k \in \mathbb{R}$).
3. **Continuous equation** (isolating k , when $v_1, v_2 \neq 0$): $(x - p_1)/v_1 = (y - p_2)/v_2$.
4. **Slope-intercept equation:** $y - p_2 = m \cdot (x - p_1)$, where $m = v_2/v_1$ is the slope.
5. **Explicit equation:** $y = m \cdot x + n$, where $n = -m \cdot p_1 + p_2$ is the y -intercept.
6. **General equation:** $A \cdot x + B \cdot y + C = 0$, with $A = v_2$, $B = -v_1$, $C = -v_2 p_1 + v_1 p_2$.

Example: line r through $P(1, 2)$ with direction $v = (1, -2)$:

- Vectorial: $(x, y) = (1, 2) + k \cdot (1, -2)$
 - Parametric: $x = 1 + k$, $y = 2 - 2k$
 - Continuous: $(x - 1)/1 = (y - 2)/(-2)$
 - Explicit: $y = -2x + 4$
 - General: $2x + y - 4 = 0$
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6. Geometric Interpretation of SLE (Rouché-Capelli in 2D and 3D)

6.1 In the plane (2 variables)

Each equation of a system with 2 unknowns represents a **line**. For m lines:

- Consistent & independent $\rightarrow$ all lines meet at exactly one point.
- Consistent & dependent $\rightarrow$ all lines coincide (same line).
- Inconsistent $\rightarrow$ no common point (for $m = 2$, the two lines are parallel and non-coincident).

6.2 In space (3 variables)

Each equation $a \cdot x + b \cdot y + c \cdot z = d$ represents a **plane** π .

- **Consistent & independent** ($\text{rk}(A) = \text{rk}(M) = 3$): the planes meet at a single point (the unique solution).
- **Consistent & dependent, 1 degree of freedom** ($\text{rk} = 2$): the planes intersect along a common **line**.
- **Consistent & dependent, 2 degrees of freedom** ($\text{rk} = 1$): the planes are all **coincident**.
- **Inconsistent**: the planes share no common point (e.g., two parallel non-coincident planes, or three planes with no common intersection even though pairs might intersect).

A **line in space** can itself be represented as the intersection of two non-coincident planes, i.e., a consistent-dependent system of 2 equations in 3 variables. Studying the relative position between a line and a plane (or between two lines) then reduces to discussing a combined 3- or 4-equation system with the Rouché-Capelli theorem: inconsistent $\Rightarrow$ parallel (no intersection); independent $\Rightarrow$ intersect at one point; dependent $\Rightarrow$ the line lies on the plane (or the two lines coincide).

7. Selected Worked Exercises (from the self-assessment sets)

Matrix equation (solve for X): given $X \cdot A = B$, isolate X by right-multiplying both sides by A^{-1} (only valid since $\det(A) \neq 0$): $X \cdot A \cdot A^{-1} = B \cdot A^{-1} \implies \mathbf{X} = \mathbf{B} \cdot \mathbf{A}^{-1}$.

Matrix polynomial: computing $A^2 - 2A$ for a given 3×3 matrix A is done by first computing the matrix product $A \cdot A$, then subtracting $2A$ entry-by-entry (both are ordinary matrix operations from Section 1.3); the exercise is a good check of matrix-multiplication mechanics, and in the solved example the result is exactly $-I_3$.

Independence via a parameter: for vectors u, v, w depending on a parameter a , build the matrix with them as rows and set its determinant, a polynomial in a , different from 0; solving that polynomial equation for its roots identifies exactly the values of a to *exclude*.

Inverse matrix with a parameter: the inverse of a matrix that depends on a parameter a exists precisely for the values of a that keep $\det(A) \neq 0$; solving $\det(A) = 0$ for a gives the excluded values, and the adjugate/transpose formula is then applied treating a as a symbolic constant.

Determinant that is always zero (property-based): a 3×3 determinant like $|\begin{bmatrix} 1, a, b+c \\ 1, b, a+c \\ 1, c, a+b \end{bmatrix}|$ can be shown to equal 0 without expanding numerically, by adding column 2 and column 3 (property 6: a column written as a sum splits the determinant), which reveals a column that is a linear combination of the others (property 7), or by producing two matrices with an identical column (property 3) whose determinants cancel.

Cramer's rule with a fixed 3×3 subsystem inside a 4-equation system: when a system has more equations than the rank warrants, remove the extra (dependent) equation, find a non-zero $r \times r$ minor, and apply Cramer's rule to the reduced $r \times r$ Cramer's system for the r "pivot" variables, leaving the rest as free parameters (used repeatedly to solve dependent 3-and-4-variable systems).

Discussion with a parameter k (Rouché–Capelli in practice): for a system whose coefficient matrix depends on k , first compute $|A|$ as a polynomial in k and factor it (often into recognizable pieces like $(k-1)^2(k+2)$); every root of that polynomial is a case to study separately (checking $\text{rk}(M)$ there), while every other value of k gives an independent system directly.